

MoCHII User Guide

Abstract

How to build and run MoCHII (MOnte Carlo for HII regions): requirements, the complete input-parameter reference, output-file formats, worked examples keyed to the regression tests, and the runbook for adding an ion. The physics and validation are documented separately in docs/MoCHII_physics.pdf; the atomic-data fitting pipeline in docs/MoCHII_fitting.pdf.

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1. Requirements and build

- Intel oneAPI MPI Fortran (mpiifort/mpiifx; GNU mpif90 supported by the Makefile), HDF5 (default prefix /data/opt/hdf5_intel; override with make HDF5_PREFIX=...), CFITSIO.
- The SEDust dust-emission library: SEDust_lib/ in this tree holds libsedust.a and its .mod files (copied from the MoCafe v2.00 build). Only needed at link time; the SEDust *data* directory is referenced at run time (Section 2.7).

- Python 3 with numpy/scipy (h5py, astropy, matplotlib, PyNeb for the test and pipeline scripts).

Build:

```
cd MoCHII
make          # -> MoCHII.x   (always a full clean rebuild)
make DEBUG=1  # bounds/traceback build
```

The Makefile intentionally performs a full rebuild every time (the MoCafe policy): there is no inter-module dependency tracking, and stale `.mod` files after editing `define.f90` otherwise cause runtime namelist errors.

Run:

```
mpirun -np 8 ./MoCHII.x input.in
```

MoCHII is MPI-only; the octree and all leaf arrays live in MPI-3 shared memory (one copy on each node).

2. Input parameters

All input is one `¶meters` namelist assigning fields of `par`. Parameters not listed keep the defaults shown. Stage switches compose: `gas_niter = 0` gives a G0 single pass (rates only); adding `gas_niter > 0` iterates the ionization (G1); `solve_te` adds the thermal balance (G2); `use_metals` the trace-metal registry; `diffuse_field` the explicit diffuse field (G3, forces case A); `refine_front` the I-front re-refinement (G4); `ion_add_dust` the dust coupling.

2.1 Photons and RNG

parameter	default	meaning
no_photons	1e5	source packets in each iteration
no_print	1e7	progress-print stride
iseed	0	RNG seed (0 = from clock/pid)

2.2 Grid and dust density

parameter	default	meaning
grid_type	'car'	'amr' (octree) or 'car' (single-level Cartesian, leaf list in raster order; no tree and no geometry arrays in memory — cell centers and half-widths are computed analytically from the raster index; refine_front unavailable). 'uniform' is accepted as an alias for 'car'
car_walk	'dda'	car-grid traversal: 'dda' (default) = dedicated incremental Amanatides–Woo walks ($t_{\max}/t_{\Delta}$ per axis, comparisons and additions only — the fastest car walk); 'shared' = the octree walks with integer index arithmetic replacing the neighbor table, a verification mode that is bit-identical to 'amr'
amr_file	''	generic AMR leaf file (FITS/HDF5/text): columns x, y, z, level, nH [, metallicity, xHI, ndust]. If empty, a 'car' grid is built from the namelist instead ('amr' requires the file)
nx, ny, nz	1, 1, 11	'car'-from-namelist cell counts (cubic cells required: $2x_{\max}/n_x = 2y_{\max}/n_y = 2z_{\max}/n_z$)
xmax, ymax, zmax	1, 1, 1	'car'-from-namelist box half-extents [distance_unit]; box is $[-x_{\max}, x_{\max}]$ etc.
nH_const	-1	if ≥ 0 , override every leaf's n_H with this uniform value [cm^{-3}] (both 'car' and 'amr'); < 0 keeps the file/column density
rmax, rmin	-999, 0	spherical density cut on leaf centers (both grids): $n_H = 0$ for $r > r_{\max}$ ($r_{\max} > 0$) and for $r < r_{\min}$ ($r_{\min} > 0$; a shell); radii in distance_unit
distance_unit	'kpc'	'kpc'/'pc'/'au'/'' (code units)
dust_model	'global_dgr'	leaf grey opacity: 'none', 'global_dgr' ($n_{H\text{cext}} \cdot \text{DGR}$), 'from_file' (ndust column), 'laursen09' (static xHI column), 'laursen09_live' (computed x_{HI} , refreshed each iteration)
cext_dust	1.6059e-21	reference extinction per H [cm^2] at lambda_ref
DGR, Z_global, Z_ref	1e-2, 0.0134, 0.0134	dust-model scalings
f_ion_dust	0.01	laursen09 survival of dust in ionized gas
f_pah, f_ion_pah	0, 0	PAH share of the reference extinction and its own ionized-gas survival (laursen09_live)

2.3 Ionizing/FUV band and the source

parameter	default	meaning
use_ion_band	F	must be T (the MoCHII transport)
nnu_ion	16	log-spaced bins; gates use 32; raise when widening the band
eion_min, eion_max	13.598, 100	ionizing band edges [eV]
add_fuv	F	FUV extension: extra log bins on [efuv_min, eion_min] below the unchanged ionizing bins; transports FUV grain heating and FUV photoionization of low-threshold metals (Mg I, C I, S I, Fe I); luminosity stays the IONIZING luminosity (Q_H preserved, the FUV part rides on top)
efuv_min	6.0	FUV lower edge [eV]
nnu_fuv	8	FUV bins (16 recommended with metals)
tstar	-999	Planck source temperature [K]
ion_spectrum	"	alternative 2-column spectrum file (E [eV], L_E arb.)
luminosity	1.0	BAND luminosity [erg/s]
xs_point, ys_point, zs_point	0	point-source position (code units, recentered box)

2.4 Initial state and iteration

parameter	default	meaning
xHI_init	1.0	initial n_{HI}/n_H (an xHI file column wins); an ionized start (1e-3) converges much faster than neutral
xHeI_init, xHeII_init	1, 0	initial He fractions
He_abund	0.1	n_{He}/n_H
gas_niter	0	iterations (0 = G0 rates-only pass)
gas_tol	1e-3	convergence tolerance on x_{HII} (measure set by conv_crit)
conv_crit	'cell'	'cell': cell-max tests ($\max \Delta x_{HII} $ — stalls at the front-cell Monte Carlo noise floor $\sim 2-4e-2$ at $4e7$ packets); 'vol': volume-integrated tests $\sum V \Delta x_{HII} / \sum V x_{HII} < gas_tol$ and $\sum V n_e \Delta T_e / \sum V n_e T_e < gas_tol_te$ (recommended; noise-robust, and the n_e weight removes vacuum and no-heating cells). The vol measures floor at their own Monte Carlo level ($\sim 3e-3$ and $7e-3$ at $8e6$ packets, scaling as $1/\sqrt{N}$) — set the tolerances just above it
ion_relax	1.0	under-relaxation on x (< 1 damps front oscillation)
case_ab	'B'	case A/B recombination ('A' forced by the diffuse field); also selects the SH95 H I/He II line tables (case-A/B)

2.5 Thermal balance and metals

parameter	default	meaning
solve_te	F	bisection on T_e coupled with the ionization
te_fixed	1e4	fixed T_e [K] when solve_te is off
te_min, te_max	1e3, 5e4	bisection bracket [K]
gas_tol_te	1e-3	convergence on $\max \Delta T_e /T_e$
atomic_dir	'data/atomic'	coefficient files (relative to the run directory)
use_metals	F	registry cascade + Tier-1 metal cooling
ci_model	'voronov'	collisional ionization: 'voronov' (default) or 'dere_hybrid' (Voronov $\times$ the Dere 2007/Voronov ratio; matters only in shielded low-ionization zones)
abund_C/N/O/Ne/S	HII20 values	$n(X)/n(H)$; an element is active when > 0
abund_Ar/Mg/Fe	0	further registry elements (gas-phase values; Mg/Fe are heavily depleted)
abund_Si/Cl/Ca	0	five-stage registry elements (gas-phase; Si and especially Ca heavily depleted, Cl nearly undepleted — Orion- like $3e-6 / 3e-7 / 2e-8$); adds [Si II] 34.8 μm , Si III/IV, [Cl II/III/IV] (5518/5538 density pair), Ca II, [Ca V]
ion_metal_abs	T	metal photoionization absorption in the band opacity (ac- tive with use_metals); the only gas opacity in the FUV bins

parameter	default	meaning
emis_output	F	write <code>_emis</code> (HDF5/FITS): leaf-by-leaf line emissivities + the state needed for maps (Section 3.)
h_coll_effects	F	add the collisional H I Balmer component (excitation of neutral H) as separate hc rows/blocks; matters at fronts and hot cells
fe_levels_full	F	load the expanded Fe II/III models (<code>nlevel_fe_{2,3}_full.txt</code> , 16/34 levels) instead of the compact 13/14 for iron-line studies
diffuse_field	F	explicit ground-recombination packets (G3)
hei_diffuse	F	fourth diffuse channel (needs <code>diffuse_field</code>): He I excited-level recombination photons (19.82 eV, 584 Å, two-photon) — correct energy bookkeeping; moves Te by $\sim 1\%$ at HII40, He split unchanged
gaunt_vh14	F	free-free Gaunt factors from the van Hoof et al. (2014) table (<code>data/gauntff_vh14.dat</code>) instead of Hummer (1988); the two agree to $\lesssim 0.5\%$ in T_e
metal_ne	F	PDR: electrons from the metal cascade join the n_e closure (the only electrons beyond the I-front)
metal_heat	F	PDR: metal photoheating in the thermal balance
grain_pe	F	PDR thermal package: grain photoelectric heating + recombination cooling (Bakes & Tielens 1994) from the local FUV field (needs <code>add_fuv</code>), plus H-impact [C II] 158 μm / [O I] 63 μm cooling; scaled by <code>pe_scale</code> and the leaf dust amount
pe_scale	1.0	photoelectric-efficiency scale factor

2.6 I-front re-refinement (G4)

parameter	default	meaning
refine_front	F	rebuild the octree on the I-front once
refine_iter	20	at which iteration
refine_lbase, refine_lmax	5, 7	base/front levels
refine_eps	1e-2	front flag: $\epsilon < x_{\text{HI}} < 1-\epsilon$
refine_dx	0.2	or a face-neighbor x_{HI} jump above this

2.7 Dust in the band and dust emission

parameter	default	meaning
ion_add_dust	F	dust absorption in the band (dusty Strömgren)
ion_dust_kext	”	kext table with EUV coverage (e.g. data/kext_albedo_WD_MW_3.1_60_D03.all_2009); required
ion_dust_scatter	F	sample dust scattering (HG, albedo weights)
dust_sed	F	SEDust stochastic dust SED at output time
dust_model_sed	'astrodust'	'astrodust'/'dl07'/'zubko'
sed_workdir	”	a SEDust sed/ tree (dielectric tables are read relative to it), e.g. the MoCafe copy .../MoCafe_v2.00/SEDust/sed
sed_qtable, sed_sizedist	MoCafe defaults	optics/size-distribution tables (relative to sed_workdir)
sed_NT, sed_Tlo, sed_Thi	200, 2.7, 5e3	SEDust temperature grid
dust_emis_bands	unset	up to 8 wavelengths [μm] (e.g. 8, 24, 70, 160): SEDust band emissivities of each leaf $\rightarrow$ _dustemis (blocks emis_dust/wl_dust [$\text{erg s}^{-1} \text{cm}^{-3} \mu\text{m}^{-1}$]); the Python map maker projects them into dust-band images (the IR is optically thin)
sed_pah_live	F	x_{HII} -dependent PAH survival in the emission: the leaf spectrum is assembled from the model channels with the PAH channel weighted by $(x_{\text{HI}} + f_{\text{ion,pah}} x_{\text{HII}})$ (needs a model with a 'PAH' channel, i.e. astrodust); turns the 8- μm map into the classic PAH ring at the ionization front

2.8 Peel-off imaging

parameter	default	meaning
ion_peel	F	after convergence, one extra transport pass peels direct (stellar + diffuse) and dust-scattered contributions toward the observers $\rightarrow$ <code>_image</code> (blocks <code>direct_ion/scatt_ion</code> , plus <code>_fuv</code> with <code>add_fuv</code>) [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$]
peel_bins	F	bin-resolved cubes <code>direct_cube/scatt_cube</code> (n_x, n_y, n_ν) in addition to the band-integrated channel images (hardness maps, synthetic EUV/FUV photometry)
obsx, obsy, obsz	unset	observer positions (or directions) in code units; or use <code>alpha/beta/gamma</code> angle lists [deg]; default: one observer on +z far away
distance	auto	observer distance (code units)
nxim, nyim	129, 129	image pixels
dxim, dyim	auto	pixel scale [deg]; auto-sized to cover the box

2.9 Output

`out_file` (base name; extension follows `file_format = 'hdf5' or 'fits'`).

3. Output files

3.1 <base>_rates.h5

One dataset for each extension, leaf-ordered:

extension	content
Gamma_HI/HeI/HeII	photoionization rates [s^{-1}]
Heat_HI/HeI/HeII	photoheating per particle [erg/s]
Heat_dust	grain heating [$\text{erg s}^{-1} \text{cm}^{-3}$] (when <code>ion_add_dust</code>)
T_dust	equilibrium mixture dust temperature [K]
x_HI, x_HeI, x_HeII, n_e, T_e	converged state
x_<el>_stages	metal ion fractions (stage, leaf)
J_nu	band mean intensity (bin, leaf) [$\text{erg/s/cm}^2/\text{Hz/sr}$]
E_bin, dE_bin	band grid [eV]
LeafXYZ	leaf centers (code units)

3.2 Text outputs

file	content
_lines.txt	line luminosities: SH95 H I and He II recombination lines (1640, 4686, ... when an He III zone exists), Porter He I lines (4471, 5876, 6678, 7065, 10830, ...) + Tier-2 collisional lines (+ hc collisional H I Balmer rows with h_coll_effects); columns elem, stage, $\lambda[\text{\AA}]$, L , $L/L(\text{H}\beta)$
_emis.h5	(with emis_output) leaf-by-leaf line emissivities: emis_h (SH95 lines) and emis_<el>_<stage> blocks [$\text{erg s}^{-1} \text{cm}^{-3}$; $\sum \text{emis} \cdot V = L$, rows match _lines.txt], wl_* wavelengths [$\text{\AA}$], plus LeafXYZ, LeafSize, n_H, n_e, T_e, x_HI/HeI/HeII, x_<el>_stages — self-contained for line maps
_nebcont.txt	nebular continuum L_ν (fb+ff tables, Milne, two-photon)
_dustir.txt	equilibrium-mixture dust IR spectrum (1–1000 μm) with the energy-closure header
_dustsed.txt	SEDust stochastic dust SED with PAH bands (when dust_sed)
_dustemis.h5	(with dust_emis_bands) SEDust band emissivities of each leaf, same block layout as _emis — readable by EmisData for dust-band maps
_image.h5	(with ion_peel) peel-off images for each observer: direct_ion/scatt_ion (+_fuv) [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$], suffix _obs<k> when several observers

3.3 Reading outputs in Python

tools/python/mochii_output.py is an importable module (works directly in a notebook) that reads every section file (HDF5 or FITS; read_sections), the line table (read_lines_table), and the text spectra (read_spectrum), and builds 2D maps from the emissivity file:

```
import sys; sys.path.insert(0, ".../MoCHII/tools/python")
import matplotlib.pyplot as plt
from mochii_output import EmisData

d = EmisData("run_emis.h5")
d.info()                # blocks, fields, line count
d.line_list()           # every stored line (block, wl)

# ready-made figures (notebook): returns (fig, ax)
fig, ax = d.plot_line("o_3", 5007.)          # log S map
fig, ax = d.plot_line("h", 6563., unit="intensity",
                      photons=True)         # photons/s/cm^2/sr
fig, ax = d.plot_field("T_e", weight="EM")
fig, axs = plt.subplots(1, 2); d.plot_line("s_2", 6717., ax=axs[0]) \
```

```
; d.plot_line("s_2", 6731., ax=axis[1]) # doublet side by side
```

```
# raw arrays, when the figure is yours to build
img, ext = d.line_map("o_3", 5007., axis="z", npix=512)
img, ext = d.line_map("h", 6563., unit="intensity", photons=True)
img, ext = d.field_map("T_e", weight="EM") # LOS-weighted average
Q = d.line_luminosity("h", 4861., photons=True) # photon rate [1/s]
```

Line maps are flux-conserving: each leaf's luminosity is deposited with exact area overlap onto the pixel grid (AMR leaf sizes handled), so $\sum SA_{\text{pix}} = L_{\text{line}}$ to machine precision. `unit='flux'` (default) gives surface brightness [$\text{erg s}^{-1} \text{cm}^{-2}$]; `unit='intensity'` gives specific intensity [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$] = $S/4\pi$ (isotropic emission, optically thin); `photons=True` converts either to photon units (divides by hc/λ of the line). A `slab=(lo,hi)` cut restricts the line of sight. Field maps average `T_e`, `n_e`, `x_HI`, ... along the line of sight with weight 'EM' ($n_e^2 V$), 'ne', 'nH', or 'V'. The same module doubles as a command-line tool for quick looks:

```
python3 mochii_output.py run_emis.h5 # list lines
python3 mochii_output.py run_emis.h5 --line o_3 5007 \
    --npix 512 --log --out o3.png
python3 mochii_output.py run_emis.h5 --line h 6563 \
    --unit intensity --photons # photons/s/cm^2/sr
python3 mochii_output.py run_emis.h5 --field T_e --weight EM
```

4. Worked examples (the regression tests)

Each tests/ directory is self-contained: grid builder or a symlinked grid, the input file, and a `check_*.py` gate script.

directory	what it demonstrates / gate
g0_gamma	rate integrals vs. analytic attenuation (bias +0.004%)
g1_stromgren	Strömgren sphere, case B fixed T_e ; AMR $\equiv$ uniform
g2a_thermal	thermal H/He nebula vs. the 1D exact reference
g2_hii	MOCASSIN Lexington HII20/HII40 benchmarks + line fluxes
g4_refine	I-front re-refinement vs. native level 7
g4_tng	Illustris-TNG post-processing demo (shadows, maps)
d_dusty	dusty Strömgren; scattering bracket; T_d/IR
peel	peel-off images vs. the optically thin analytic flux (+0.017%); bin cubes
uni_dda	uniform grid: shared walk $\equiv$ octree (bit-identical); DDA walk to rounding
pdr	FUV/PDR switches: carbon-electron shell, photoelectric T_e

A minimal thermal + metals + dust input (from `d_dusty`):

```
&parameters
par%no_photons = 4.0e7 par%iseed = 1234
```

```

par%grid_type = 'amr'           par%amr_file = 'amr_L6.fits'
par%distance_unit = 'pc'       par%dust_model = 'global_dgr'
par%use_ion_band = .true.      par%nnu_ion = 32
par%eion_min = 13.598         par%eion_max = 100.0
par%tstar = 4.0e4             par%luminosity = 3.178e38
par%xHI_init = 1.0e-3         par%He_abund = 0.1
par%gas_niter = 60            par%gas_tol = 2.0e-3
par%solve_te = .true.         par%atomic_dir = '../data/atomic'
par%use_metals = .true.
par%ion_add_dust = .true.
par%ion_dust_kext = '../data/kext_albedo_WD_MW_3.1_60_D03.all_2009'
par%cext_dust = 4.868e-22      par%DGR = 1.0
par%out_file = 'run.h5'        par%file_format = 'hdf5'
/

```

5. Adding an ion (the G5 runbook)

Adding an ion touches only data:

1. tools/fitting/fit_cooling_tier1.py: add the ion with a T_i ladder guessed from its .elvlc level energies; run; check the reported max error and the overlay figure.
2. tools/fitting/fit_nlevel_tier2.py: add the ion with the level count needed for its diagnostics; run.
3. tools/fitting/make_element_data.py: add the element to ELEMENTS (Z, stage count) and its thresholds; charge exchange entries where sources exist; run. The CHIANTI .rrparams/.drparams types 1–3 are handled (RR/RR2/DR/DR2 lines).
4. Fortran: a par%abund_<El> field and one entry in the species_setup name/abundance lists (only for a NEW element; new stages of a known element need nothing).
5. Verify: extend verify_nlevel_pyneb.py with a diagnostic ratio where PyNeb has data; run a smoke test and check the new lines in _lines.txt.

Data caveats met so far: Ar II has no CHIANTI v11 level/collision data (stage tracked without lines); Fe III collision strengths differ from PyNeb's BB14 by up to ~16% in 4658/4702 (known spread for iron); charge-exchange stage labels differ between the Huang (2023) and MOCASSIN transcriptions for Mg/Fe (the Huang labels are adopted; see make_element_data.py).

6. Practical notes

- **Bins:** 32 bins give $\sim 2\%$ rate-integral discretization; use 64 for sub-percent work. The `add_fuv` extension carries its own bins below `eion_min`, so the ionizing resolution is unaffected.
- **FUV and the PDR:** without `add_fuv` there is no radiation beyond the I-front and the metals there recombine toward neutral; with it, FUV penetrates the neutral gas (dust extinction only) and produces the PDR-side C II and Mg II. Metal photoheating and grain photoelectric heating are not in the thermal balance, so PDR-zone temperatures are indicative only.
- **Convergence:** use `conv_crit = 'vol'` for production — the cell-max criterion stalls at the front-cell noise floor (and, with `add_fuv`, at no-heating cells where T_e pins to `te_min`). Both measures are printed each iteration. The G3 diffuse field smooths the front and converges far better.
- **Starts:** prefer an ionized initial state; a neutral start advances the front $\sim$ one cell per iteration.
- **kext tables:** must cover the band (the `astrodust` table stops at $912 \text{ \AA}$; use the D03 table for EUV work). Tables are sorted on read (the D03 file has an out-of-order seam).
- **Re-refinement:** one event per run; the old shared-memory windows are reclaimed only at exit.
- **SEDust:** `sed_workdir` must point at a SEDust `sed/` tree; the exact stochastic solve costs $\sim 0.05\text{--}0.1$ s for each gas leaf (distributed over ranks).